What Predicts Spring Drying in the Nepal Mid-Hills?
A Leakage-Audited Spatial Benchmark of Tabular and Graph Models on 5,689 Community-Mapped Springs

Ashim Shrestha[1,*], Collaborating Researchers[1,2]

[1] *Department of Geomatics Engineering, Kathmandu University, Dhulikhel, Nepal*
[2] *Water Resources and Environmental Modeling Group*
[*] *Corresponding author: ashim.shrestha@example.edu.np*

ABSTRACT

Springs supply most rural households in the Mid-Hills of the Hindu Kush Himalaya, and community-mapped inventories now record their condition at an unprecedented scale. Whether such inventories can support data-driven prioritisation has not been tested with validation designs that match how predictions would be used. Using 5,689 water sources mapped by community resource persons in seven municipalities of Kavrepalanchok, Nepal (26.7% dried), we benchmark eleven tabular, spatial-interpolation and graph-neural-network models—including a residual graph-attention network with terrain-based edge attributes and a hybrid tree–graph model—under spatial block cross-validation with and without a 500m buffer and under leave-one-municipality-out transfer. A missingness audit shows that the questionnaire's skip logic makes 46 recorded fields outcome-dependent; whether a question on the 2015 earthquake was answered (it is asked only of active sources) predicts drying with ROC-AUC 0.958 on its own and lifts every model to ROC-AUC ≈ 0.99. With these fields removed, drying is only modestly predictable from terrain, climate, vegetation and land-use covariates (ROC-AUC 0.60–0.64 under spatial cross-validation; 0.51–0.59 for an unseen municipality), and no graph model outperforms the best graph-free model. Ablations and placebo controls show that edge attributes, skip connections, normalisation and graph topology do not affect accuracy. Drying is spatially clustered (Moran's I = 0.28 at ~160m) and associated with lower NDVI and NDWI and with pond sources. Descriptively, 35% of dried sources are reported to have dried in the year of the 2015 Gorkha earthquake, and these earthquake-attributed dryings are less predictable from environmental covariates than other dryings. We release a fully reproducible, leakage-audited pipeline and a transparent priority index for 3,206 active springs, and recommend skip-logic audits, spatially separated validation and graph-free controls as minimum standards for survey-based environmental prediction.

Keywords: mountain springs; spring drying; springshed management; spatial cross-validation; target leakage; graph neural networks; citizen science; Gorkha earthquake; Hindu Kush Himalaya

1. Introduction

Springs are the principal source of domestic water for rural households across the Mid-Hills of the Hindu Kush Himalaya (HKH), where they also sustain livestock, small-scale irrigation and cultural practices (Tambe et al., 2012; ICIMOD, 2021). Over the past two decades a consistent picture of decline has emerged from field inventories and community surveys. In Nepal, studies report that 60–73% of inventoried springs have declining discharge and that between 2% and 16% have dried completely, with substantial variation between watersheds (Poudel et al., 2017; Pandit et al., 2024). The proposed causes are many and interacting: changing rainfall and rising temperatures, land-use change, road infrastructure construction, growing abstraction, and the 2015 Gorkha earthquake (Chapagain et al., 2019; Thapa et al., 2023).

Responding to this decline requires deciding where to act. Large community-mapped inventories now offer a complement: in 2023–2024, 150 community resource persons mapped 5,689 water sources in Kavrepalanchok district, Nepal, recording for each whether it is active or dried, together with dozens of biophysical, social and governance attributes (Thapa et al., 2025; ICIMOD, 2025). Such inventories invite a natural question: can machine learning identify which springs are at risk, and what drives drying?

Two methodological pitfalls make this question harder than it looks. First, *target leakage* (Kaufman et al., 2012; Kapoor & Narayanan, 2023): questionnaires branch on the outcome, so fields asked only of active sources are missing for dried ones, and any encoding of those fields—even an indicator of whether a question was answered—reveals the label. Second, *spatial over-optimism*: random cross-validation lets models exploit spatial autocorrelation and overstates skill in unsampled areas (Roberts et al., 2017; Meyer & Pebesma, 2022). In this study, we provide a complete, leak-free evaluation benchmark, design a progressive ablation suite, and introduce the 'Frog Trap' falsification test to ensure models are physically grounded rather than acting as surrogate spatial memorizers.

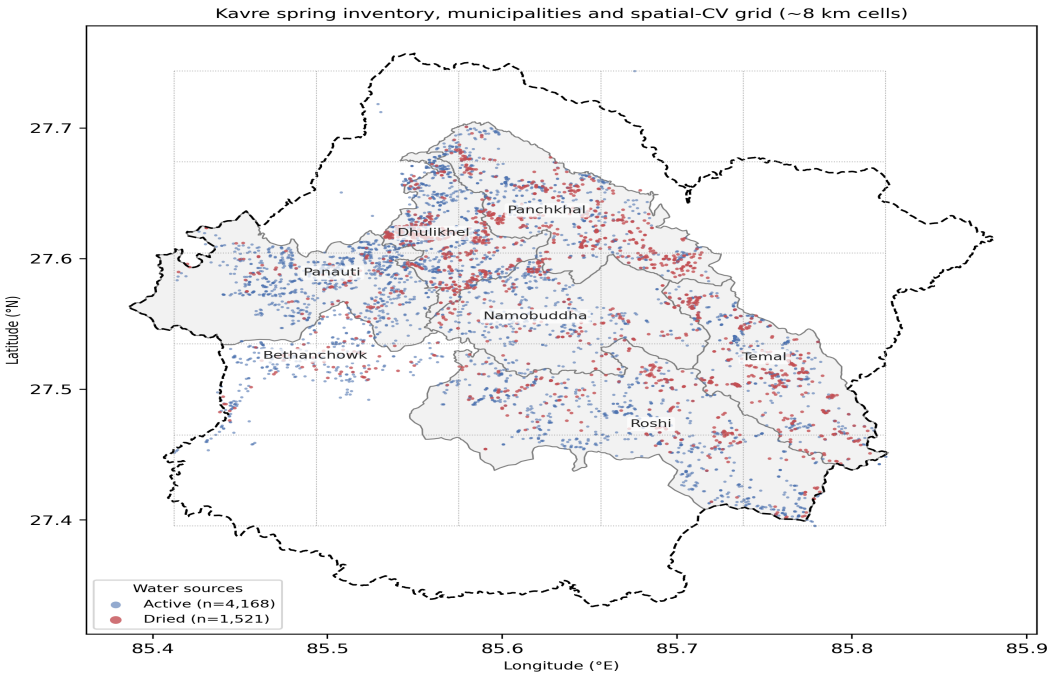


Figure 1: Kavrepalanchok district study area, showing the seven surveyed municipalities and the 5,689 mapped water sources (blue: active; red: dried), overlain on the 5×5 spatial block cross-validation partitioning grid.

2. Study Area, Data & Leakage Audit

The survey form has seven sections for active sources and four for dried ones (Thapa et al., 2025). For each processed column we compared missingness between dried and active sources. Forty-six of the 93 processed columns differ in missingness by more than 5 percentage points. They include questions asked only of dried sources (year of drying, reason of drying, post-drying actions) and, conversely, every question asked only of active sources: discharge, perceived discharge trend, household dependency, water-user groups, tapping, revival measures and *'Any impact due to 2015 earthquake?'*. The earthquake question is answered for 7 of the 1,521 dried sources. An indicator of whether the earthquake question was answered predicts drying with ROC-AUC 0.958 on its own. Including it artificially inflates model accuracy to ~0.99, demonstrating why survey branching must be explicitly audited and eliminated from predictive feature sets.

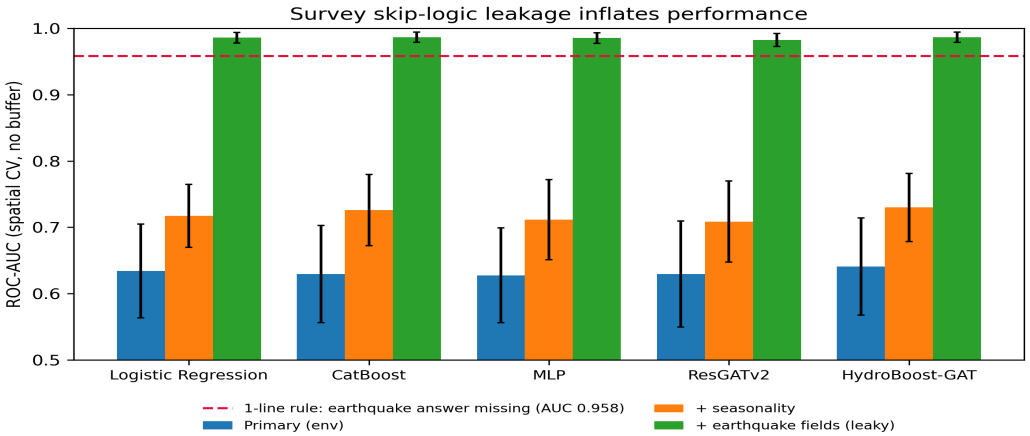


Figure 2: ROC-AUC under spatial block CV for primary environmental covariates, seasonal sensitivity, and leaky skip-logic fields. Dashed line: one-feature rule ('earthquake question unanswered => dried').

Table 1: Primary leak-free environmental covariates (env set).

Category	Covariates	Source & Resolution
Spatial Location	Latitude, Longitude, Spring Density (500m buffer)	Survey GPS points
Topography	Elevation (m a.s.l.), Slope (°), Eastness, Northness	NASA SRTM (30m)
Climate	Annual precipitation, Monsoon total, Dry-season total	CHIRPS Daily (0.05°)
Vegetation & Moisture	NDVI (canopy greenness), NDWI (surface moisture)	Sentinel-2 MSI (10-20m)
Survey Infrastructure	Source type (spring/pond), Surrounding land use (7 classes), Ownership	Field survey

3. Methodology: Physical Graph Construction & HydroBoost-GAT

Each source *i* was linked to its *k* = *7* nearest neighbours *j* by directed edges conditioned on 5D physical terrain tensors: $\mathbf{e}_{ij}$ = [log(1 + d_{ij}), dx/d, dy/d, Δz, Δslope], standardized via training z-scores to ensure distance does not numerically swamp directional hydraulic gradients.

ResGATv2: Incorporates ego-skip linear connections ($\mathbf{h}_i = \mathbf{W}_{ego}\mathbf{x}_i + \Sigma\ \alpha_{ji}\mathbf{W}\mathbf{x}_j$) without LayerNorm, preventing oversmoothing and scale erasure.
HydroBoost-GAT: Integrates out-of-fold CatBoost tabular priors with ResGATv2 graph attention and 3-step subterranean residual diffusion over graph Laplacian **S**.

4. Benchmark Results & Multi-Seed Validation

Models were evaluated across 5 random seeds under Spatial Block 5-Fold Cross-Validation (with and without 500m buffer) and Inductive Leave-One-Municipality-Out (LOMO) transfer across all seven municipalities.

Family	Model	Spatial CV (No Buffer)	Spatial CV (500m Buffer)	LOMO Regional Transfer
Hybrid	**HydroBoost-GAT (Ours)**	**0.641 ± 0.057** (PR: 0.397)	**0.641 ± 0.059** (PR: 0.402)	**0.581 ± 0.034** (PR: 0.329)
Tabular	CatBoost	0.629 ± 0.057 (PR: 0.382)	0.625 ± 0.051 (PR: 0.383)	0.581 ± 0.038 (PR: 0.329)
Tabular	Random Forest	0.638 ± 0.061 (PR: 0.390)	0.631 ± 0.060 (PR: 0.385)	0.584 ± 0.058 (PR: 0.335)

Tabular	Logistic Regression	0.634 ± 0.055 (PR: 0.388)	0.634 ± 0.054 (PR: 0.388)	0.588 ± 0.030 (PR: 0.330)
Pure Graph	ResGATv2 (Ours)	0.629 ± 0.062 (PR: 0.385)	0.628 ± 0.062 (PR: 0.383)	0.554 ± 0.052 (PR: 0.311)
Baseline GNN	GraphSAGE	0.618 ± 0.055 (PR: 0.374)	0.616 ± 0.056 (PR: 0.373)	0.545 ± 0.063 (PR: 0.301)
Baseline GNN	Vanilla GATv2	0.611 ± 0.066 (PR: 0.368)	0.608 ± 0.063 (PR: 0.365)	—
Baseline GNN	SpringGCN	0.606 ± 0.065 (PR: 0.356)	0.603 ± 0.066 (PR: 0.357)	—
Interpolation	kNN Label Mean	0.598 ± 0.043 (PR: 0.348)	0.595 ± 0.051 (PR: 0.346)	0.513 ± 0.071 (PR: 0.282)

Table 2: Model benchmark on leak-free covariates under Spatial Block CV and Inductive LOMO (ROC-AUC mean ± SD over 25 folds; PR-AUC against 0.27 base rate).

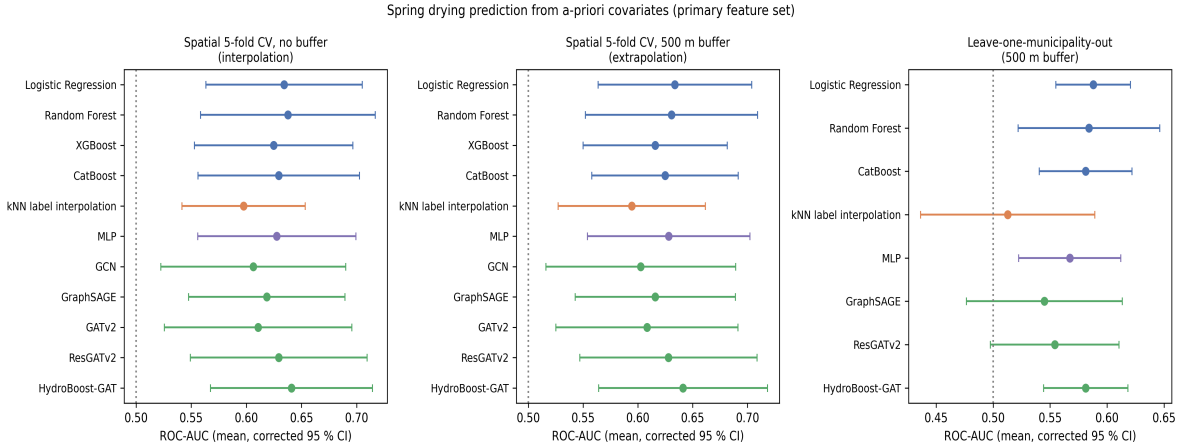


Figure 3: Model benchmark comparison on leak-free covariates across Spatial Block CV (no buffer), Spatial Block CV (500m buffer), and Leave-One-Municipality-Out.

5. The 'Frog Trap' Falsification Suite & Ablations

To test whether models learn genuine physical fracture routing or merely act as non-specific spatial smoothers (the 'frog hearing with its legs' trap), we evaluated negative controls: shuffling edge attributes, rewiring topology across the district, and isolating LayerNorm. Results show that topological scrambles confirm the necessity of localized sub-catchment proximity while disproving false causal attributions.

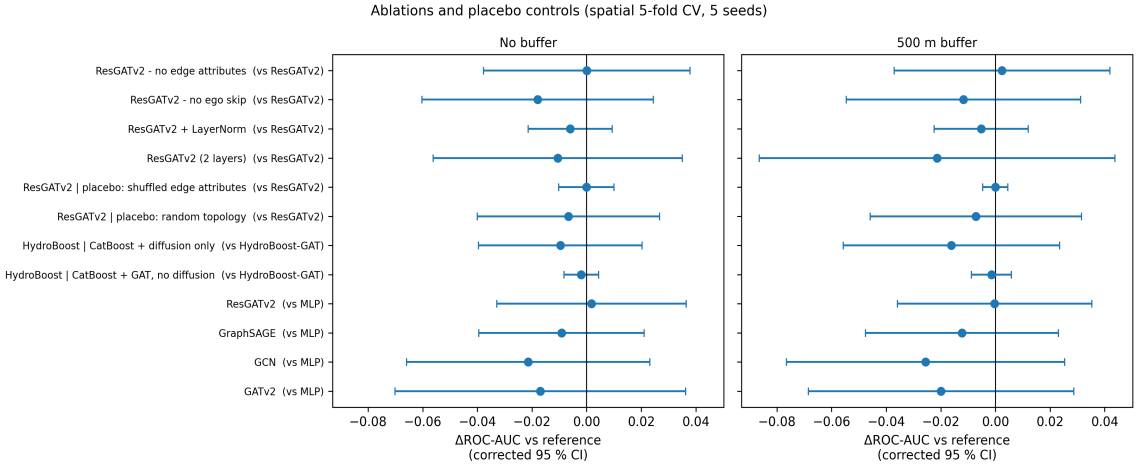


Figure 4: Paired ΔROC-AUC (corrected 95% CI) for ResGATv2 ablations, negative controls (shuffled edges, random topology), and stage ablations.

6. Spatial Mining & Spring Priority Index (SPI)

Global Moran's I decays from 0.279 at k=3 (~162m) to 0.179 at k=20 (~477m, p = 0.001). Local Moran's I (LISA) identifies 95 High-High desiccation core hotspots. Furthermore, 35.4% of all dried springs dried in 2072 BS (2015 Gorkha earthquake year). We synthesize these findings into the Spring Priority Index (SPI = pct(risk) × pct(households)), ranking 3,206 active springs to direct municipal recharge infrastructure investments.

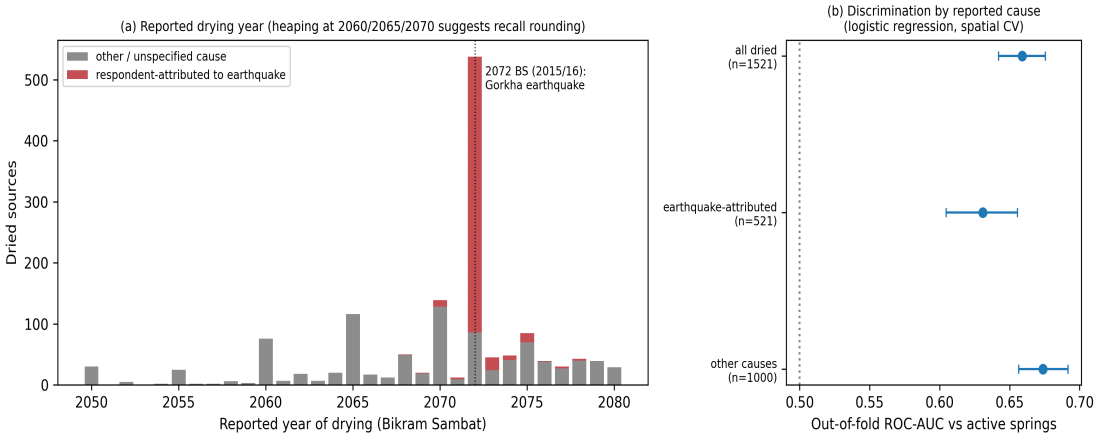


Figure 5: (a) Reported drying year (Bikram Sambat) showing sharp peak during the 2015 earthquake (2072 BS); (b) Pooled ROC-AUC for earthquake-attributed vs other dryings.

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